SULIXAY VILAIPHONE（罗意）

Nationality: Laos | **Date of Birth:** 2001/11/20 | **WeChat:** luoyo321 | **Email:** Sulixay2001@gmail.com

GitHub: https://github.com/luoyi-321
Hugging face: https://huggingface.co/LuoYiSULIXAY

EDUCATION background

Xihua University | Master's Degree in Computer Science and Technology (Academic Program)
2022 – 2025

- Awarded the Chengdu Government Scholarship.
- Honored as an Outstanding Volunteer of Pidu District.
- University-level Third Prize in the "Internet Plus" Innovation and Entrepreneurship Competition
- Research focus: Artificial Intelligence – Wireless Sensing.
- CGPA: 3.38/4.00

Xihua University | Bachelor's Degree in Computer Science and Technology
2020 – 2022

- Won "The Best Ideas" Award at the National Startup Competition of Laos.
- Outstanding Graduate of 2022.
- CGPA: 3.93/4.00

Sichuan Vocational and Technical College | Associate Degree in Computer Application Technology

- Awarded the **Sichuan Provincial Government Scholarship**.
- University-level Third Prize in the "Internet Plus" Innovation and Entrepreneurship Competition

WORK / PROJECT EXPERIENCE

- **Cropchai: AI-Powered Crop Disease Detection and Treatment Application | 2025 – Present**
  Built an intelligent mobile application leveraging computer vision and deep learning to detect crop diseases with real-time image analysis. Integrated treatment recommendation engine that provides farmers with actionable insights and remediation strategies. Designed user-friendly interface to make AI diagnostics accessible to agricultural users with varying technical backgrounds.
- **Lao Language Speech Recognition and Synthesis System Development | 2025 – Present**
  Developed end-to-end speech processing solutions for Lao language, implementing both speech-to-text and text-to-speech modules. Leveraged advanced neural network architectures and acoustic modeling to achieve high accuracy in tone-sensitive language processing. Focused on optimizing performance for real-world deployment with resource constraints.
- **SAR Echo Generation Software on GPU (Core Algorithm Developer) | June 2024 – March 2025**
  Developed real-time SAR echo modulation coefficient calculation and simulation on a GPU platform. The real-time SAR modulation module generates echoes, clutter, and interference by convolving with the modulation coefficients while considering platform motion characteristics.
- **Conventional Echo Generation Software on GPU (Algorithm Developer) | April 2024 – January 2025**
  Developed GPU-based generation of echo, clutter, and interference signals for virtual-real collaborative modulation systems.
- **Wireless Non-Line-of-Sight (NLOS) System Development (Core Algorithm Developer) | June 2023 – April 2024**
  Developed an interactive platform for human posture estimation using WiFi devices and deep learning on CSI signals for through-wall sensing. Designed two experience-based games: a shooting game and a motion game.
- **Advisor for 2024 Project | 2024**
  Provided technical and strategic guidance for a key project in 2024. Supported project planning, system design, and solution validation, ensuring high-quality outcomes and smooth project progression.
- **President, International Student Union of Xihua University** | **2023 – 2025**
  Led the International Student Union, overseeing event planning, student affairs, and multicultural communication. Coordinated cross-department activities to enhance campus engagement for over 100 international students. Strengthened organizational structure and improved administrative efficiency.
  Additional projects, experiments, and open-source contributions available at GitHub and Hugging Face.

RESEARCH ACHIEVEMENTS

- **1st Author: *"WiFi Person Pose Estimation via Interpretable Convolutional Modular for Person Spatial Map" (IEEE Signal Processing Letters, SCI Q2, IF: 3.89,)***
  This study proposes a WiFi-based human pose estimation method using channel state information (CSI), integrating subcarrier correlation with a UNet architecture. By constructing a high-resolution spatial correlation map, the method achieves precise keypoint localization and significantly improves mPCK performance in through-wall environments.
- **1st Author: "SSEM-LSEM Enhanced Multi-Space Fusion: Cost-Efficient Pose Estimation from WiFi Signals" (IEEE Internet of Things, IF: 8.9, Revise and Resubmit)**
  Proposed a WiFi-based indoor pose estimation system using CSI signals and UPA antenna arrays. Designed a dual-stream neural network (SSEM-LSEM) that processes signal amplitude and 2D-DOA features for efficient human pose detection without line-of-sight requirements.

- **2nd Author: "Utilizing Baidu Index to Investigate the Public Concern Towards Vocational Education in the Chinese Mainland"** 。
  This study uses Baidu Index big data to analyze public attention toward "vocational education" in mainland China from 2014 to 2024, examining temporal trends, geographic distribution, and user characteristics. The study reveals the direct impact of policy events on public interest, providing data support and analytical paradigms for policy evaluation and social cognition research.
- **3rdAuthor:** ***Patent: "A Real-Time Through-Wall Human State Estimation Method Based on WiFi Devices"*** **(Application No. 202411866951.7)** 。
  This invention proposes a real-time through-wall human posture estimation method using WiFi devices, including system deployment, data acquisition, preprocessing, and pose estimation.

SKILLS

Coding skills:

- c++/c、Java、python、web、dart、sql。

Software Skill

- vsCode\NetBean IDE\MatLab\flutter\figma。
- Office\LaText\Photoshop\coreldraw \video editor 。

Frameworks and Libraries：

- **Wireless Sensing & Signal Processing:** WiFi CSI–based human posture estimation, UWB/Radar data processing, time–frequency analysis, DOA estimation.
- **Deep Learning & AI**: Model training and finetuning (CNN, RNN, Transformer), pose estimation, speech synthesis (F5-TTS), OCR, and computer vision tasks.
- **GPU & High-Performance Computing:** CUDA programming, kernel optimization, shared memory usage, parallel computing for SAR echo generation.
- **Computer Vision:** YOLO-based detection, CRNN/LPR models, OpenCV image processing, tracking and segmentation.
- **Speech & NLP:** Training personalized neural codec models, Lao language TTS/ASR dataset creation and model adaptation.
- **Backend & Deployment:** FastAPI / Flask model inference services, REST API deployment, server configuration, inference endpoint optimization.
- **Tools & Dev Skills:** Git/GitHub, Linux, Docker, FFmpeg, MATLAB, Pandas, NumPy, Scikit-learn.

LANGUAGE SKILLS：

- Chinese（HSK6+）、Thai（Fluent）、English（good）、Lao（Mother tongue）。